



Webserverlidar.ino

```
1 #include <WiFi.h>
2 #include <AsyncTCP.h>
3 #include <ESPAsyncWebServer.h>
4 #include <LDS_YDLIDAR_X2_X2L.h>
5
6 // Replace with your network credentials
7 const char* ssid = "ESP32-LiDAR-AP";
8 const char* password = "12345678";
9
10 // Create AsyncWebServer object on port 80
11 AsyncWebServer server(80);
```

Fig. 5: Code snippet for configuring the AP SSID and password

Let's review the lidar datasheet. The YD X2 lidar provides rotation angle and distance data via serial peripherals at a baud rate of 115200. It includes a connector board for USB connection to a PC (see Fig. 3 and Table 2).

As shown in Table 3, the lidar senses distances from 0.12 to 8 metres,

scanning 360 degrees at 3000Hz. With this data, we can now connect the lidar to the IndusBoard. You can power both components with a 5V battery or an adaptor. For a portable solution, use the battery; otherwise, the 5V adaptor works well. Fig. 3 shows the lidar connector pins.

Circuit diagram

Fig. 4 shows the circuit diagram for the IndusBoard standalone IoT lidar radar. The device is powered by a 5V

Webserverlidar.ino

```
26 <title>ESP32-S2 LiDAR Map</title>
27 <style>
28   body { font-family: Arial, sans-serif; text-align: center; margin-top: 50px; }
29   .card { display: inline-block; margin: 20px; padding: 20px; border: 1px solid #ccc; border-radius: 10px; }
30   canvas { margin-top: 20px; }
31 </style>
32 </head>
33 <body>
34   <h1>Real-Time LiDAR Map</h1>
35   <div class="card">
36     <h2>Angle</h2>
37     <p id="angle">0°</p>
38   </div>
39   <div class="card">
40     <h2>Distance</h2>
41     <p id="distance">0 mm</p>
42   </div>
43   <canvas id="lidarMap" width="400" height="400"></canvas>
44   <script>
45     const ctx = document.getElementById('lidarMap').getContext('2d');
46
47     function drawMap(angle, distance) {
48       ctx.clearRect(0, 0, ctx.canvas.width, ctx.canvas.height);
49       ctx.beginPath();
50       const radian = angle * (Math.PI / 180);
51       const x = (ctx.canvas.width / 2) + (distance * Math.cos(radian)) / 10;
52       const y = (ctx.canvas.height / 2) + (distance * Math.sin(radian)) / 10;
53       ctx.moveTo(ctx.canvas.width / 2, ctx.canvas.height / 2);
54       ctx.lineTo(x, y);
55       ctx.stroke();
56     }
57
58     function updateData() {
59       fetch('/data')
60         .then(response => response.json())
61         .then(data => {
62           document.getElementById('angle').textContent = data.angle + '°';
63           document.getElementById('distance').textContent = data.distance + ' mm';
64           drawMap(data.angle, data.distance);
65         })
66         .catch(error => console.error('Error:', error));
67     }
68     updateData();
69   </script>
70 </body>
71 </html>
```

Fig. 6: Web page's code snippet